

Wave equation for a 2D membrane (FEM version)

May 26, 2026

1 the problem

We solve the advection-diffusion equation

$$\frac{\partial u}{\partial t} = -v_x \frac{\partial u}{\partial x} - v_y \frac{\partial u}{\partial y} + D \nabla^2 u, \quad (1)$$

with the velocity field (v_x, v_y) and the diffusion constant independent of the position. We use a square computational box $(x, y) \in (0, L) \times (0, L)$ and the finite element method and introduce a periodic boundary conditions

$$u(x + mL, y + nL) = u(x, y) \quad (2)$$

for integer m and n . In FEM application of the periodic boundary conditions is easy: the nodes from the edges of the box should be attributed the same global number, e.g. the node at the upper edge (x, L) gets the same global number as the node in the position $(x, 0)$ etc.

In our previous assignments we had evaluated the global matrix of $-\nabla^2$ (the stiffness matrix **S**). We need to evaluate the matrix of the advection operator. For a constant velocity the local matrix is the same for all the elements. The local matrix for the first derivative in x direction is evaluated as

$$\left(\frac{\partial}{\partial x}\right)_{ji} = \frac{a}{2} \sum_l \sum_n w_l w_n g_j(p_l, p_n) \frac{g_i(p_l + \Delta, p_n) - g_i(p_l - \Delta, p_n)}{2\Delta}.$$

We evaluate similarly $\left(\frac{\partial}{\partial y}\right)$ matrix and the assembled global matrix of the advection operator is denoted by **C**, and the overlap matrix is denoted by **O**

The Crank-Nicolson step is calculated as the solution of the linear system of equations

$$\left(\mathbf{O} - \frac{\Delta t}{2}(\mathbf{C} - \mathbf{S})\right) \mathbf{c}(t + \Delta t) = \left(\mathbf{O} + \frac{\Delta t}{2}(\mathbf{C} - \mathbf{S})\right) \mathbf{c}(t).$$

Lets set $\Delta t = 0.02$, $L = 5$ and the initial condition: $u(x, y) = \exp(-2(x - L/2)^2 - 2(y - L/2)^2)$.

2 Deliverables

- pure advection $D = 0$, $v_x = v_y = 1$. Solve the equations till the center of the density returns to its initial position second time. Is its shape constant?
- pure diffusion $D = 0.1$, $v_x = v_y = 0$. Plot the minimal and maximal values of u at the nodes as a function of time.
- advection diffusion $D = 0.1$, $v_x = v_y = 1$. Is the velocity of the center of the packet equal to the velocity field vector?